Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/29/2020
PI Name:	Venkateswaran, Kasthuri Ph.D.		
Project Title:	Omics in Space (OIS): Technology Development for Omics Instrumentations and Biomolecule Measurements		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017 HERO NNJ16ZSA001N-TRIRT. Appendix C: Translational Research Institute for Space Health (TRISH) Research Topics
Start Date:	10/01/2017	**End Date:**	09/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2019 (original end date was 9/30/2022) per E. Urquieta/TRISH (Ed., 9/4/19) NOTE: End date changed to 6/30/2019 (original end date was 9/30/2022) per E. Urquieta/TRISH (Ed., 5/23/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Landry, Kyle Ph.D. (Liberty Biosecurity) Mohan, Ganesh Babu Malli Ph.D. (NASA Jet Propulsion Laboratory) Mason, Christopher Ph.D. (Weill Medical College of Cornell University) Regberg, Aaron Ph.D. (NASA Johnson Space Center) Singh, Nitin Ph.D. (NASA Jet Propulsion Laboratory) Smith, David Ph.D. (NASA Ames Research Center) Urbaniak, Camilla Ph.D. (NASA Jet Propulsion Laboratory) Weislogel, Mark Ph.D. (IRPI) Sinclair, David Ph.D. (Institutional PI: Harvard College Medical School) Wong, Season Ph.D. (Institutional PI: AI Biosciences, Inc.)		
Grant/Contract No.:	NNX16AO69A-T0101		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The omics in space (OIS) project will develop instrumentations for extracting nucleic acids (NA) and sequencing methodologies for inflight detection and measurement of several biomolecules related to physiological and immunological effects related to spaceflight (e.g., aging, crew health, elevated antimicrobial resistance, and virulence). Onboard the International Space Station (ISS), we will generate data related to microbiome (targeted for bacteria, fungi, and viruses) to address the microbial composition, metagenome (crew and environments) for functional characteristics, miRNA for immunological function, as well as epigenetic data to elucidate radiation-induced damage, cellular disruption, and aging of astronauts. The next generation sequencing (NGS) using the MinION platform and PCR instruments (SmartCyler and RAZOR) were adopted by NASA last year and were successful in sequencing libraries that had been prepared on Earth and shipped to the ISS. Limitation in this study was that samples had to be prepared on Earth before sequencing could be performed on the ISS, as there are no technologies in place to process samples, extract biomolecules, and prepare libraries for NGS sequencing. The OIS team recognizes these limitations and will be developing an automated Sample Processing Instrumentation for NA extraction that is streamlined, requires minimal crew time, reducing the amount of contamination between samples, and producing consistent results. The OIS team will generate a vast amount of data utilizing molecular methods in space to provide the NASA Translational Research Institute (NTRI) and Human Research Program with an opportunity to understand the omics changes under microgravity and microbial communities of crew and habitat. The state-of-the-art molecular methods will allow researchers to not only describe the changes in microbial communities of ISS, but also distinguish whether these biological signatures are of any concern to crew health and engineering systems. Focused and targeted molecular approaches are likely to reveal a subset of novel and medically important microbes posing particular threats to crew health. Archiving of omics data and their availability for extensive data mining may feed into functional genomics activities. It is expected that a range of metadata will allow NASA to correlate microbial community information with temporal or environmental changes. The development of an all-encompassing, integrated OIS database will enable various phylogenetic- and pathogenic-based strategies of screening for, and identifying, specific subsets of microorganisms (e.g., dominating viral and microbial pathogens, as well as those that bear resistance traits relevant to antibiotics). This dataset will (a) create a capability for NASA to compare fluctuating microbial communities to “baseline” standards; (b) enable more accurate assessments of crew health associated with mission planning; (c) allow evidence-based development of future bio-load management policies, particularly for long duration missions; and (d) capitalize on parallel research from non-NASA institutions efforts. The mission of the NTRI is to lead a national effort to apply innovative terrestrial research to spaceflight human risk mitigation strategies for long-duration exploration missions. The OIS project will target biomolecules that may be responsible for physiological and immunological conditions that affect humans during spaceflight missions. The OIS project is a consortium of investigators from various NASA centers, industries, and academia to solve microgravity-related sample processing issues. This is a four-year TRI Core Research Project (TCore), consisting of an interdisciplinary team of omics researchers, experts in biomolecule assay development, and clinicians. Three science principal investigators (PIs) will lead the team and ensure that everyone works in a coordinated fashion.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	REPORTING SUBMITTED JANUARY 2019 International Space Station Medical Project (ISSMP) assessment on OIS project on the feasibility for the flight definition is completed. NASA Institutional Review Board (IRB) report recommended to fund OIS project for Year 1 to develop Sample Processing Instrument (SPI) for flight. The ground validated SPI is already available and ready to be tested for the flight compatibility. Since the ground validated SPI is already available and ready to be tested for the flight compatibility, the maturation of the SPI will help with extracting nucleic acids in space.
	FINAL REPORTING SUBMITTED DECEMBER 2019 Results: a. The OIS team validated an automated nucleic acid extraction device for omics in space using whole cell microbial reference standards during Year 1. NASA has made great strides in the last five years to develop a suite of instruments for the International Space Station in order to perform molecular biology in space. However, a key piece of equipment that has been lacking is an instrument that can extract nucleic acids from an array of complex human and environmental samples. The Omics in Space team has developed the µTitan (simulated micro(µ) gravity tested instrument for automated nucleic acid)) system capable of automated, streamlined, nucleic acid extraction that is adapted for use under microgravity. The µTitan system was validated using a whole cell microbial reference (WCMR) standard comprised of nine bacteria titrated at concentrations that would challenge the performance of the instrument as well as resolve detection limits for isolating DNA. Quantitative assessment of system performance was measured by comparing instrument input challenge dose vs recovery by Qubit spectrofluorometry, RTqPCR, Bioanalyzer, and next generation sequencing. Overall, results indicate that the µTitan system performs equal to or greater than a similar commercially available, Earth-based, automated nucleic acid extraction device. The µTitan system was also tested in Yellowstone National Park (YNP) with the WCMR, to mimic a remote setting, with limited resources. The performance of the device at YNP was comparable to that in a laboratory setting. Such a portable, field deployable, nucleic extraction system will be valuable for environmental microbiology as well as health care diagnostics in preventing microbial diseases. ANNUAL REPORTING SUBMITTED JANUARY 2019 (1) Original project aims/objectives: Objective: Instrument development to process samples in space (AI Biosciences, Inc., IRPI LLC, Jet Propulsion Laboratory); 20% of effort

Task Progress:	The MinION on the ISS has the ability to revolutionize science in space by allowing real time sequencing of NA from a variety of samples. Although sequencing has been achieved in space recently, sample processing is still a challenge due to lack of microgravity executable methods and or tools. The proposed OIS project will develop microgravity-compatible instrumentations to allow for (i) extracting nucleic acids; (ii) amplifying target gene(s) using PCR; and (iii) preparing genomic libraries suitable for the MinION sequencer. Task 1. Design and assemble a working NA extraction prototype cartridge for PCR and NGS Task 2. Demonstrate efficiency of the prototype to extract NA in space from a variety of samples Task 3. Confirm the integrity and purity of the NA for downstream analyses for PCR and NGS in space Task 4. Establish device performance analysis for ISS applications (2) Key findings: a. The ISSMP and NASA IRB decided that Sample Processing Instrument should be developed before flight feasibility is assessed. Also, ISSMP is requesting OIS team to reduce the crew time for this project. b. During the flight definition period, the OIS-project team did not receive any funds to carry out the above tasks. (3) Impact of key findings on hypotheses, technology requirements, objectives, and specific aims of the original proposal: Worked with ISSMP for the feasibility of the OIS flight definition phase. The OIS team is in dialogue with the ISSMP and the Translational Research Institute for Space Health (TRISH) to request appropriate crew time. (4) Results: It is decided that in Year 1, OIS-team will develop the Sample Processing Instrument (SPI) that is flight worthy before considering the feasibility of the project for flight.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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